

**MACHINE LEARNING-BASED STUDENT PERFORMANCE
AND DROPOUT RISK PREDICTION FOR EARLY
IDENTIFICATION OF AT-RISK STUDENTS**

PHASE II REPORT

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in partial fulfillment for the award of the degree
of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



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OCTOBER 2026

ANNA UNIVERSITY, CHENNAI
BONAFIDE CERTIFICATE

Certified that this project report titled “**MACHINE LEARNING-BASED STUDENT PERFORMANCE AND DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-RISK STUDENTS**” is the bonafide work of **MUSA HAJI (230701391), PRITHIVI RAJ (230701508), SACHIN CHAUDHARY (230701402)** who carried out the project work under my supervision. Certified further that, to the best of my knowledge, the work herein does not form part of any thesis or dissertation on the basis of which degree or award was conferred on an earlier occasion on this or any other candidates.

“This project addresses the following Sustainable Development Goal(s):

Goal - 4 : Quality Education; Goal - 10 : Reduced Inequalities.”

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ABSTRACT

With the growing access to student academic and behavioral data, it is now possible to identify early which students are likely to face academic challenges. Traditional assessment relies on manual grading and attendance tracking, which does little to uncover complex relationships between academic, demographic, and behavioral characteristics. This project develops a machine learning-based system that predicts student performance and supports the early identification of at-risk students. After data preprocessing, six classifiers, namely Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, and Gradient Boosting, are evaluated using stratified cross-validation and hyperparameter optimization. The optimized Gradient Boosting model performed best, achieving 87.34% accuracy, 87.95% precision, 87.34% recall, and an F1-score of 87.42%. SHAP-based explainability shows that prior academic performance, especially the G2 grade, contributes most to the predictions. Although the dataset has no dropout-labelled target, the framework provides a foundation for a dedicated dropout-risk prediction system once labelled dropout data becomes available.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This work presented the design and implementation of a machine learning-based framework for predicting student academic performance and supporting the early identification of at-risk students using academic, personal, and behavioral data.

Six classification models, namely Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, and Gradient Boosting, were evaluated using 5-fold stratified cross-validation and Grid Search hyperparameter tuning. The optimized Gradient Boosting model achieved the highest cross-validation F1-score of 0.8891 and was selected as the final model. On the test set it obtained an accuracy of 87.34%, precision of 87.95%, recall of 87.34%, and an F1-score of 87.42%.

The integration of SHAP-based explainability makes the predictions transparent. The analysis shows that previous academic performance, especially the G2 grade, has the strongest influence on the predicted outcome. This allows educators to understand why a student is flagged and to plan timely, targeted support.

6.2 FUTURE WORK

The most important enhancement is a separate dropout-risk prediction module trained on a labelled dataset containing past dropout outcomes, so that students can be classified by their likelihood of dropping out. The system can also be extended with real-time attendance, assignment submission, and learning-management-system data for continuous monitoring and earlier intervention.